```
import pandas as pd
df = pd.read_csv('/content/Social_Network_Ads.csv')
df.head()
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

Next steps: Generate code with df New interactive sheet

```
X = df.iloc[:, 2:4].values   # Age, Estimated Salary
y = df.iloc[:, 4].values     # Purchased

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)

print(f"Shape of X_train: {X_train.shape}")
print(f"Shape of X_test: {X_test.shape}")
print(f"Shape of y_train: {y_train.shape}")
print(f"Shape of y_test: {y_test.shape}")
```

```
Shape of X_train: (300, 2)
Shape of X_test: (100, 2)
Shape of y_train: (300,)
Shape of y_test: (100,)
```

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
model.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression()
```

```
y_pred = model.predict(X_test)
print(y_pred)
```

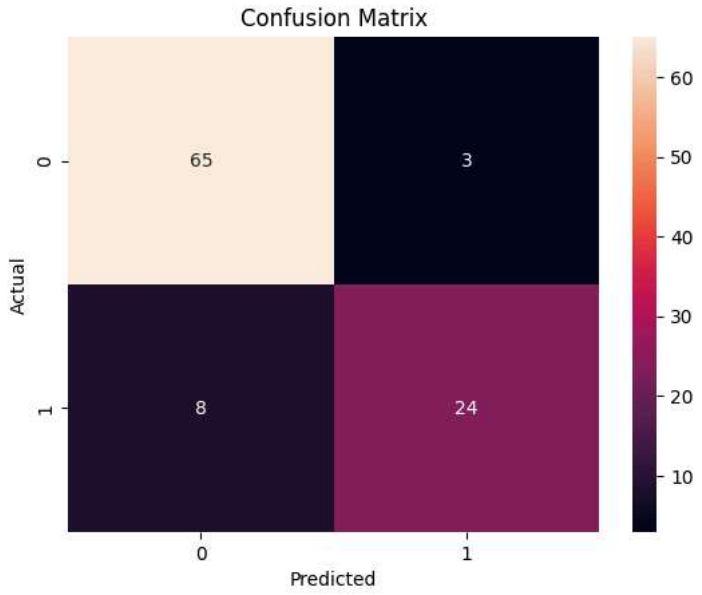
```
[0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 0 1 0 0 1 0 1 0 1 0 0 0 0 0 0 1 0 0 0 0
 0 0 1 0 0 0 0 1 0 0 1 0 1 1 0 0 0 1 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 1 0 0 0
 0 0 1 0 1 1 1 1 0 0 1 1 0 1 0 0 0 1 0 0 0 0 0 0 1 1]
```

```
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)

import seaborn as sns
import matplotlib.pyplot as plt
plt.figure()
sns.heatmap(cm, annot=True, fmt="d")
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# Extract TP, FP, TN, FN
TN = cm[0][0]
FP = cm[0][1]
FN = cm[1][0]
TP = cm[1][1]

print("\nTP:", TP)
print("FP:", FP)
print("TN:", TN)
print("FN:", FN)
```




```
TP: 24
FP: 3
TN: 65
FN: 8
```

```
# Accuracy
accuracy = (TP + TN) / (TP + TN + FP + FN)

# Error Rate
error_rate = (FP + FN) / (TP + TN + FP + FN)

# Precision
precision = TP / (TP + FP)

# Recall
recall = TP / (TP + FN)

print("\nAccuracy:", accuracy)
print("Error Rate:", error_rate)
print("Precision:", precision)
print("Recall:", recall)
```

```
Accuracy: 0.89
Error Rate: 0.11
Precision: 0.8888888888888888
Recall: 0.75
```